USING AI FOR FRAUD DETECTION IN UPI PAYMENTS

ENSEMBLE LEARNING FOR UPI FRAUD RISK ASSESSMENT

A Review-Based Research Paper

Research Topic: Artificial Intelligence for UPI Payment Fraud Detection

Paper Type: Literature Review and Conceptual Analysis

Scope: Fraud detection methods, machine learning approaches, practical challenges, privacy, and future scope

Keywords: UPI, Artificial Intelligence, Machine Learning, Fraud Detection, Ensemble learning, Digital Payments, Risk Assessment

ABSTRACT

One model may be strong at known fraud but weak at novel behaviour. An anomaly detector may surface new patterns but generate too many false alerts. Combining models can improve coverage, yet it may also increase cost and reduce interpretability. To examine the conceptual value of ensembles and hybrid models for UPI fraud-risk assessment. This review synthesizes relevant concepts and published financial-fraud research. It presents a conceptual analysis rather than original experiments. No live UPI records are accessed and no model is claimed to have been implemented.

Keywords: UPI, Artificial Intelligence, Machine Learning, Fraud Detection, Ensemble learning, Digital Payments, Risk Assessment

1. INTRODUCTION

Digital payment systems have changed how individuals and businesses transfer money. UPI supports interoperable transfers through participating banks and applications and is commonly used for person-to-person and merchant payments. The speed and convenience of this ecosystem make it important to protect users against fraud while preserving a smooth payment experience.

Payment fraud may involve technical compromise, but many incidents depend on social engineering. A person can be persuaded to approve a payment, trust an impersonator, scan a misleading QR code, or disclose sensitive information. A payment may therefore be technically authorized while still being induced by deception. This distinction means that fraud monitoring should be part of a broader security and customer-protection process.

One model may be strong at known fraud but weak at novel behaviour. An anomaly detector may surface new patterns but generate too many false alerts. Combining models can improve coverage, yet it may also increase cost and reduce interpretability.

AI and machine learning can assist by learning patterns from historical examples or identifying behaviour that differs from a baseline. Their predictions are not certainty: data may be incomplete, labels may be delayed, and legitimate transactions can look unusual. Human review and established security controls remain important.

2. BACKGROUND AND CONTEXT

A typical payment journey can involve a payer, a payee or merchant, an application interface, participating banks, and payment-system routing. The exact process and data available depend on the payment type and the participating institutions. A student research project should not assume access to internal bank or network data.

Potential monitoring fields may include transaction amount, timestamp, status, transaction type, protected account identifiers, and authorized session or behavioural indicators. Access should be limited to what is needed for a defined purpose. Data protection is not an optional extra: transaction histories can reveal sensitive patterns about a person's routine and relationships.

Fraud labels require a clear definition. A failed transaction may be caused by network or technical problems, while a successful payment may still be the result of deception. Reliable analysis therefore needs clear labelling rules and documentation of uncertainty.

3. PROBLEM STATEMENT

One model may be strong at known fraud but weak at novel behaviour. An anomaly detector may surface new patterns but generate too many false alerts. Combining models can improve coverage, yet it may also increase cost and reduce interpretability.

4. AIM AND OBJECTIVES

Aim: To examine the conceptual value of ensembles and hybrid models for UPI fraud-risk assessment.

1. Explain the importance of ensemble learning in the context of UPI fraud monitoring.
2. Review relevant financial-fraud and UPI-focused literature.
3. Discuss strengths, limitations, and practical design trade-offs.
4. Identify evaluation, privacy, security, and governance requirements.
5. Propose directions for future academic study without claiming implementation.

5. LITERATURE REVIEW

Ryman-Tubb, Krause, and Garn (2018) surveyed AI and machine-learning research in payment-card fraud detection and discussed the gap between research results and industry practice. Although card payments differ from UPI, the review is useful for understanding the importance of operational evaluation and deployment constraints.

Al-Hashedi and Magalingam (2021) reviewed data-mining techniques used in financial fraud detection from 2009 to 2019. Their review helps situate classification and data-mining methods within a wider financial-fraud research landscape.

Hernandez Aros and colleagues (2024) reviewed machine-learning techniques for financial fraud detection and described model families, data sources, and evaluation concerns. Such broad reviews can inform a UPI-focused conceptual paper, but findings should not be assumed to transfer directly across payment systems.

UPI-focused papers by Murkute, Jadhav, and Patil (2025), Kumar and Rani (2025), and Rayaan and Chaitra A. L. (2025) are directly related to the topic and discuss ML or deep-learning approaches for UPI transaction fraud. Their bibliographic details and full-text findings should be verified against publisher records before submission.

Across this literature, recurring concerns include rare fraud labels, changing behaviour, data quality, evaluation design, privacy, and the difference between offline results and a live payment environment. Reported accuracy values should not be compared unless the datasets, splits, labels, and metrics are sufficiently comparable.

6. TECHNICAL REVIEW

Bagging and Random Forest

Bagging trains models on resampled datasets and combines their outputs. Random Forest applies this idea to decision trees and introduces additional randomness. It can reduce variance compared with a single tree.

Boosting

Boosting adds learners sequentially, with later learners focusing on errors made earlier. It can be effective on structured transaction data but can overfit noisy labels and requires careful tuning.

Voting Ensembles

A voting ensemble combines predictions from multiple classifiers. Hard voting uses class labels, while soft voting combines probability estimates. Probability calibration matters because uncalibrated outputs can distort the combined score.

Rule-and-Model Hybrids

A hybrid may apply established security rules, then combine a supervised risk score with anomaly indicators. Each component should have a clear purpose and its contribution should be measured.

7. CONCEPTUAL EXAMPLE

A conceptual system could use a rule for a known prohibited pattern, a classifier for patterns similar to past fraud, and an anomaly detector for unusual behaviour. The combined result could trigger an appropriate response. Whether this is better than a single model must be demonstrated using comparable evaluation.

This example is illustrative only. It is not a real customer record or a measured result. Actual decisions would require authorized information, validated policies, and consideration of the customer's context.

8. PROPOSED CONCEPTUAL METHODOLOGY

Step 1: Define the use case

Define what counts as fraud, what decision the system supports, what level of intervention is permitted, and which errors are most costly.

Step 2: Select authorized data

Use an approved, anonymized dataset or carefully documented synthetic data. Record the source, date range, fields, missingness, and known limitations.

Step 3: Prepare features

Validate fields, handle missing values, protect identifiers, and calculate features using only information available at decision time. Avoid duplicates and data leakage.

Step 4: Establish a baseline

Begin with simple rules or a transparent baseline before testing more complex models. Document model settings and thresholds.

Step 5: Evaluate fairly

Use training, validation, and test sets with a suitable time-aware split. Report precision, recall, F1-score, confusion matrix, false-positive rate, and latency where relevant.

Step 6: Review and monitor

Consider human review, alert workload, privacy safeguards, model drift, and a controlled process for any future model update.

The methodology above is proposed for future work. This paper does not claim that the student group performed these steps, trained a model, connected to a payment service, or obtained experimental results.

9. DISCUSSION

The central theme of this paper is ensemble learning. Its potential value depends on whether it solves a clearly defined problem better than existing controls. A more complex technique should not be selected solely because it is newer or appears more advanced.

A responsible fraud-detection workflow balances security with usability. Missed fraud can harm customers, while false alerts can delay genuine payments and create unnecessary investigation work. The threshold and response policy should reflect the actual operating context and should be reviewed as conditions change.

Privacy and explainability are equally important. Data collection should be minimized, access should be controlled, and risk scores should be communicated as indicators rather than proof. Consequential actions require documented governance and suitable human oversight.

For an academic review project without coding, the contribution is the structured analysis of published work, the comparison of methods, and the identification of requirements for future research. The paper should not present hypothetical examples as experimental findings.

10. LIMITATIONS

This paper is a literature-based conceptual review and does not report original experiments, a trained model, or a live integration. Published studies may use different datasets and fraud definitions, so their reported metrics may not be directly comparable. Some UPI-related studies may rely on synthetic data or limited descriptions; such data can support controlled demonstrations but may not capture real-world behaviour. The paper's conclusions should therefore be understood as a structured review and proposed framework, not proof of operational effectiveness.

11. PRIVACY, SECURITY AND ETHICAL CONSIDERATIONS

Data minimization

Collect and retain only the information required for a defined purpose. Protect identifiers and restrict access according to role.

Security controls

Use secure storage and transmission, access logs, retention limits, incident-response procedures, and secure deletion.

Fairness

Monitor error patterns across appropriate transaction contexts and review whether false alerts disproportionately affect particular users or circumstances.

Human oversight

Do not treat a score as proof of intent. Use documented review procedures and provide ways to correct mistakes.

Responsible student research

Do not access private banking data without authorization or include identifiable transaction examples in reports.

12. FUTURE SCOPE

Future work on ensemble learning could begin with an authorized dataset or a clearly described synthetic dataset. The study should define the fraud label, document data limitations, and compare the proposed approach with a simple baseline.

Evaluation should use a chronological split where appropriate and report more than accuracy. Precision, recall, F1-score, false-positive rate, PR-AUC where suitable, and processing latency can provide a more complete picture. Any claimed improvement should be reproducible and evaluated on a held-out test set.

Future research should also examine robustness to changing fraud behaviour, the quality of explanations, privacy safeguards, and the operational burden of alerts. Deployment should proceed only after security, governance, and authorized data-access requirements are satisfied.

13. CONCLUSION

This review examined ensemble learning as a potential component of AI-assisted fraud detection in UPI payments. The literature suggests that machine-learning methods can support risk assessment, but their value depends on reliable data, suitable evaluation, a proportionate response process, and ongoing monitoring. Unusual activity is not automatically fraudulent, and no model can guarantee detection of every scam. A responsible approach combines analytical methods with authentication, user awareness, reporting, investigation, privacy protection, and human oversight. This paper is intended as a review-based foundation for the SPI seminar and report; it does not claim that the group coded or experimentally validated a system.

14. REFERENCES

6. Murkute, A., Jadhav, D., & Patil, Y. (2025). UPI Fraud Detection Using Machine Learning. International Journal of Creative Research Thoughts, 13(4), a651–a658. https://ijcrt.org/viewpaperforall.php?paper=IJCRT2504084
7. Kumar, J., & Rani, N. (2025). Optimized Machine Learning and Deep Learning Approaches for Effective Detection of Fraud in Unified Payments Interface (UPI) Transactions. IJSAT, 16(4). https://doi.org/10.71097/IJSAT.v16.i4.9525
8. Rayaan, S., & Chaitra, A. L. (2025). A Novice Approach for UPI Fraud Detection by Using Machine Learning Techniques. 2025 IEEE 7th International Conference on Computing, Communication and Automation. https://doi.org/10.1109/ICCCA66364.2025.11325722
9. Ryman-Tubb, N. F., Krause, P., & Garn, W. (2018). How Artificial Intelligence and machine learning research impacts payment card fraud detection: A survey and industry benchmark. Engineering Applications of Artificial Intelligence, 76, 130–157. https://doi.org/10.1016/j.engappai.2018.07.008
10. Al-Hashedi, K. G., & Magalingam, P. (2021). Financial fraud detection applying data mining techniques: A comprehensive review from 2009 to 2019. Computer Science Review. https://doi.org/10.1016/j.cosrev.2021.100402

11. Hernandez Aros, L., Bustamante Molano, L. X., Gutierrez-Portela, F., et al. (2024). Financial fraud detection through the application of machine learning techniques: A literature review. Humanities and Social Sciences Communications, 11, 1130. https://doi.org/10.1057/s41599-024-03606-0
12. Ngai, E. W. T., Hu, Y., Wong, Y. H., Chen, Y., & Sun, X. (2011). The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. Decision Support Systems, 50(3), 559–569. https://doi.org/10.1016/j.dss.2010.08.006
13. National Payments Corporation of India (NPCI). UPI: Official information and resources. https://www.npci.org.in/
14. Reserve Bank of India (RBI). Official website and payment-system publications. https://www.rbi.org.in/

Academic note: Verify exact bibliographic details, publication status, page numbers, and full-text findings through publisher records before final submission. References are provided as reading leads and should not be cited for claims that are not supported by the source.

15. EXTENDED DISCUSSION AND DESIGN CONSIDERATIONS

Balancing Security and Convenience

Payment security is not measured only by the number of alerts. A system that blocks too many genuine transactions may lead users to abandon the service or ignore warnings. Conversely, a system that minimizes friction too aggressively may allow preventable losses. The design objective is to reduce expected harm while keeping legitimate payments usable. This requires transparent thresholds, monitoring, and review of both missed cases and false alerts.

Importance of Context

Transaction amount, time, recipient novelty, and frequency can be useful signals, but their meaning depends on context. A new recipient may be a family member, a first-time merchant, or a fraudulent account. A large transfer may be a normal rent or education payment. Combining signals can be more informative than relying on a single fixed threshold, yet the model must remain open to uncertainty.

Data Limitations

Real payment data can be difficult to obtain because of confidentiality, privacy, and security obligations. Synthetic datasets offer a safer environment for learning and reproducible demonstrations, but their assumptions must be stated. A model tested on synthetic data should not be described as validated for live UPI traffic.

Human Factors

Many scams exploit urgency, fear, authority, or trust. AI can help prioritize some unusual transactions, but it cannot reliably infer a user's intention from transaction fields alone. Clear warnings, accessible reporting, safe authentication, and trained investigators remain essential.

Governance and Accountability

A deployed system should have an owner, documented purpose, approved data sources, performance monitoring, change control, and a rollback plan. Investigators should be able to see relevant alert information, record outcomes, and escalate uncertain cases. The organisation should be able to explain why a policy action was taken.

Research Reproducibility

A useful academic comparison documents the data source, time period, fraud definition, preprocessing, feature list, train/validation/test split, model parameters, threshold, and metrics. Without this information, results cannot be interpreted fairly or reproduced. The same evaluation process should be applied to every compared model.

16. SUMMARY OF KEY FINDINGS

- AI is best treated as a decision-support layer within a wider security process.
- No algorithm is universally best; model selection depends on data, fraud patterns, latency, and the cost of errors.
- Precision, recall, false-positive burden, and latency are essential alongside accuracy.
- Data quality, label delay, class imbalance, and leakage can distort apparent performance.
- Privacy, fairness, explainability, and human oversight are requirements for responsible use.
- Any implementation or empirical claim must be supported by authorized data and reproducible evaluation.